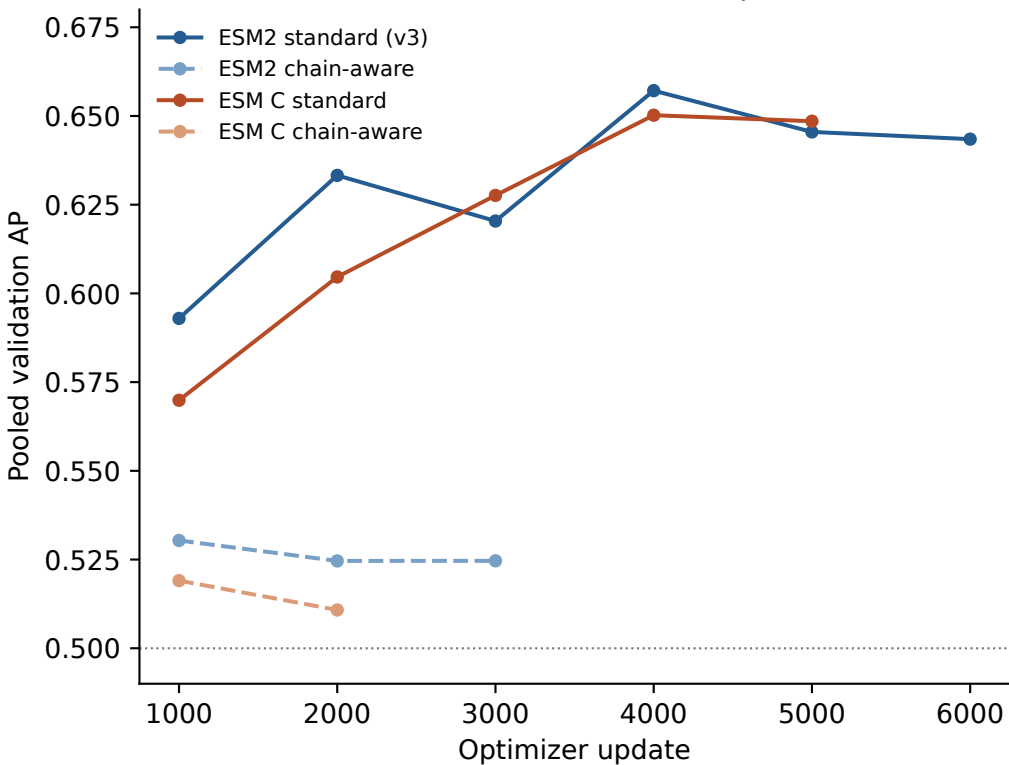
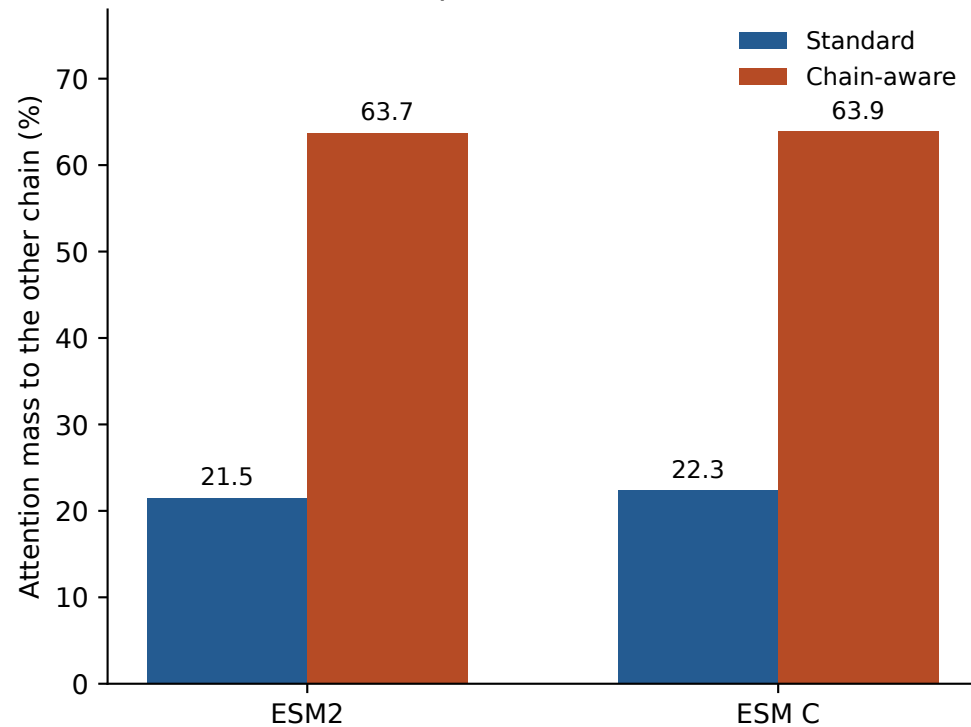


V4 chain-aware attention investigation — 1 October 2026

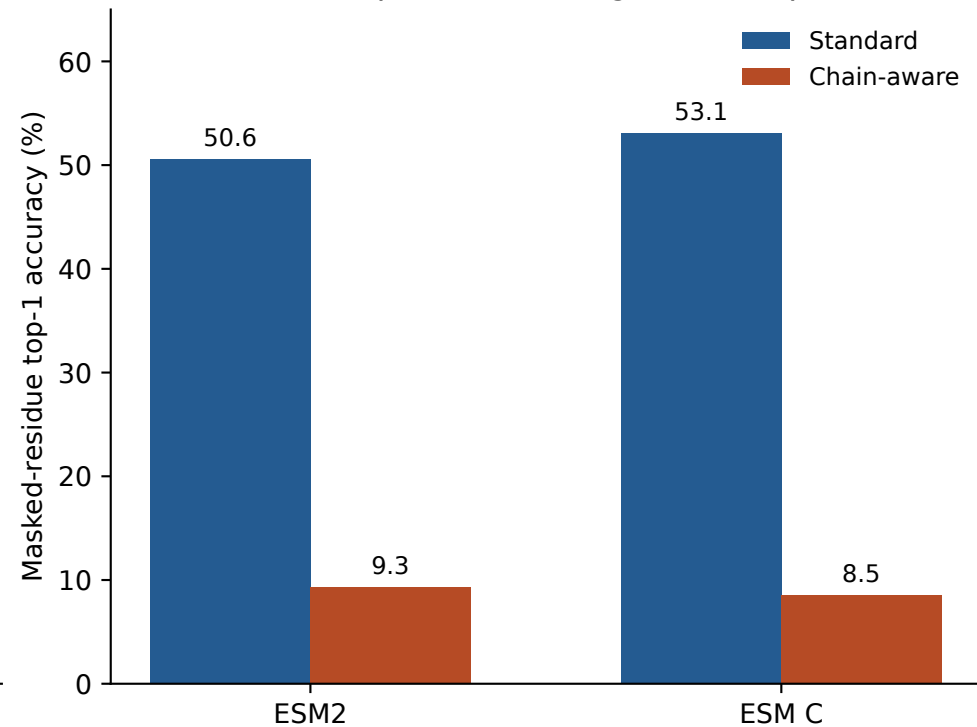
A Full validation: 59,258 pairs



B Same pretrained activations



C Same pretrained weights and inputs



B-C: eight training pairs, both orientations; C: 2,354 masked residues. Diagnostic interventions, no optimizer steps.